# 第2次实验实验报告

## 实验目的

1.针对图像分类任务，学习并实现卷积神经网络。
2.针对时序预测任务，学习并实现循环神经网络。
3.通过调整学学习率，优化方法和Batch大小，了解超参数对深度学习模型性能的影响。

## 实验内容

**一、图像分类**
对Cifar10数据集，用两种深度学习模型VGG和ResNet进行分类，对比分类结果。
在示例代码基础上增加层数，调整每个模型的训练参数，包括优化方法、学习率、batch大小等等，观察这些参数对模型训练和测试结果的影响。
**二、时序数据预测**
对UCR的ECG数据集或者其它时序数据集，用两种深度学习模型进行多步预测，对比预测结果
在示例代码基础上进行修改，并调整每个模型的训练参数，包括优化方法、学习率、时间步大小等等，观察这些参数对模型训练和测试结果的影响

## 实验步骤与结果

### 图像分类

- **定义数据集类与数据加载器**

> 对数据预处理归一化以及封装成batch

```python
class CIFAR10DataLoader:
    """CIFAR-10数据加载器类，统一管理数据加载、预处理和统计量计算"""

    def __init__(self, data_path, batch_size=64, num_workers=2, download=False):
        self.data_path = data_path
        self.batch_size = batch_size
        self.num_workers = num_workers

        # 计算数据集的均值和标准差（基于原始训练数据）
        self.mean, self.std = self._calculate_dataset_stats(download)

        # 创建数据加载器
        self.train_loader = None
        self.test_loader = None
        self._create_loaders()

        # 类别名称
        self.classes = ('plane', 'car', 'bird', 'cat', 'deer',
                        'dog', 'frog', 'horse', 'ship', 'truck')

    def _calculate_dataset_stats(self, download):
        """计算原始数据集的均值和标准差"""
        print("正在计算数据集的均值和标准差...")

        # 直接用transforms.ToTensor()加载原始数据
        temp_transform = transforms.Compose([transforms.ToTensor()])
        temp_dataset = torchvision.datasets.CIFAR10(
            root=self.data_path, train=True, download=download, transform=temp_transform
        )
        temp_loader = DataLoader(temp_dataset, batch_size=64, shuffle=False)

        channels_sum = 0
        channels_squared_sum = 0
        num_batches = 0

        for images, _ in temp_loader:
            channels_sum += torch.mean(images, dim=[0, 2, 3])
            channels_squared_sum += torch.mean(images ** 2, dim=[0, 2, 3])
            num_batches += 1

        mean = channels_sum / num_batches
        std = (channels_squared_sum / num_batches - mean ** 2) ** 0.5
```

```
        print(f"计算得到的均值: {mean}")
        print(f"计算得到的标准差: {std}")

        return mean, std

    def _create_loaders(self):
        """创建训练和测试数据加载器"""
        # 训练数据增强
        transform_train = transforms.Compose([
            transforms.RandomCrop(32, padding=4),
            transforms.RandomHorizontalFlip(),
            transforms.ToTensor(),
            transforms.Normalize(self.mean, self.std)
        ])

        # 测试数据预处理（无增强）
        transform_test = transforms.Compose([
            transforms.ToTensor(),
            transforms.Normalize(self.mean, self.std)
        ])

        # 直接创建数据集，不需要额外的Dataset类
        train_dataset = torchvision.datasets.CIFAR10(
            root=self.data_path, train=True, transform=transform_train, download=True
        )
        test_dataset = torchvision.datasets.CIFAR10(
            root=self.data_path, train=False, transform=transform_test, download=True
        )

        # 创建数据加载器
        self.train_loader = DataLoader(
            train_dataset, batch_size=self.batch_size, shuffle=True,
            num_workers=self.num_workers, pin_memory=True if torch.cuda.is_available() else Fals
        )
        self.test_loader = DataLoader(
            test_dataset, batch_size=self.batch_size, shuffle=False,
            num_workers=self.num_workers, pin_memory=True if torch.cuda.is_available() else Fals
        )

    def get_train_loader(self):
        return self.train_loader

    def get_test_loader(self):
```

```
        return self.test_loader

    def set_batch_size(self, batch_size):
        """动态修改batch size"""
        if self.batch_size != batch_size:
            self.batch_size = batch_size
            self._create_loaders()
            print(f"Batch size 已更新为: {batch_size}")

    def get_stats(self):
        """返回数据的均值和标准差"""
        return self.mean, self.std
# 类别名称
classes = ('plane', 'car', 'bird', 'cat', 'deer',
           'dog', 'frog', 'horse', 'ship', 'truck')
```

- **VGG实现**

```python
class VGG(nn.Module):
    def __init__(self, num_classes=10):
        super(VGG, self).__init__()

        # VGG11 配置
        self.features = nn.Sequential(
            # Block 1
            nn.Conv2d(3, 64, kernel_size=3, padding=1),
            nn.BatchNorm2d(64),
            nn.ReLU(inplace=True),
            nn.MaxPool2d(kernel_size=2, stride=2),

            # Block 2
            nn.Conv2d(64, 128, kernel_size=3, padding=1),
            nn.BatchNorm2d(128),
            nn.ReLU(inplace=True),
            nn.MaxPool2d(kernel_size=2, stride=2),

            # Block 3
            nn.Conv2d(128, 256, kernel_size=3, padding=1),
            nn.BatchNorm2d(256),
            nn.ReLU(inplace=True),
            nn.Conv2d(256, 256, kernel_size=3, padding=1),
            nn.BatchNorm2d(256),
            nn.ReLU(inplace=True),
            nn.MaxPool2d(kernel_size=2, stride=2),

            # Block 4
            nn.Conv2d(256, 512, kernel_size=3, padding=1),
            nn.BatchNorm2d(512),
            nn.ReLU(inplace=True),
            nn.Conv2d(512, 512, kernel_size=3, padding=1),
            nn.BatchNorm2d(512),
            nn.ReLU(inplace=True),
            nn.MaxPool2d(kernel_size=2, stride=2),

            # Block 5
            nn.Conv2d(512, 512, kernel_size=3, padding=1),
            nn.BatchNorm2d(512),
            nn.ReLU(inplace=True),
            nn.Conv2d(512, 512, kernel_size=3, padding=1),
            nn.BatchNorm2d(512),
            nn.ReLU(inplace=True),
```

```
            nn.MaxPool2d(kernel_size=2, stride=2),
        )

        self.classifier = nn.Sequential(
            nn.Linear(512 * 1 * 1, 4096),
            nn.ReLU(inplace=True),
            nn.Dropout(0.5),
            nn.Linear(4096, 4096),
            nn.ReLU(inplace=True),
            nn.Dropout(0.5),
            nn.Linear(4096, num_classes),
        )

    def forward(self, x):
        x = self.features(x)
        x = x.view(x.size(0), -1)
        x = self.classifier(x)
        return x

# 创建 VGG 模型
vgg_model = VGG(num_classes=10).to(device)
print(vgg_model)
```

- **ResNet18实现**

```python
class BasicBlock(nn.Module):
    expansion = 1

    def __init__(self, in_channels, out_channels, stride=1):
        super(BasicBlock, self).__init__()

        self.conv1 = nn.Conv2d(in_channels, out_channels, kernel_size=3, stride=stride, padding=
        self.bn1 = nn.BatchNorm2d(out_channels)
        self.conv2 = nn.Conv2d(out_channels, out_channels, kernel_size=3, stride=1, padding=1, b
        self.bn2 = nn.BatchNorm2d(out_channels)

        self.shortcut = nn.Sequential()
        if stride != 1 or in_channels != out_channels:
            self.shortcut = nn.Sequential(
                nn.Conv2d(in_channels, out_channels, kernel_size=1, stride=stride, bias=False),
                nn.BatchNorm2d(out_channels)
            )

    def forward(self, x):
        out = torch.relu(self.bn1(self.conv1(x)))
        out = self.bn2(self.conv2(out))
        out += self.shortcut(x)
        out = torch.relu(out)
        return out

class ResNet(nn.Module):
    def __init__(self, block, num_blocks, num_classes=10):
        super(ResNet, self).__init__()
        self.in_channels = 64

        self.conv1 = nn.Conv2d(3, 64, kernel_size=3, stride=1, padding=1, bias=False)
        self.bn1 = nn.BatchNorm2d(64)

        self.layer1 = self._make_layer(block, 64, num_blocks[0], stride=1)
        self.layer2 = self._make_layer(block, 128, num_blocks[1], stride=2)
        self.layer3 = self._make_layer(block, 256, num_blocks[2], stride=2)
        self.layer4 = self._make_layer(block, 512, num_blocks[3], stride=2)

        self.linear = nn.Linear(512 * block.expansion, num_classes)

    def _make_layer(self, block, out_channels, num_blocks, stride):
        strides = [stride] + [1] * (num_blocks - 1)
        layers = []
```

```
        for stride in strides:
            layers.append(block(self.in_channels, out_channels, stride))
            self.in_channels = out_channels * block.expansion
        return nn.Sequential(*layers)

    def forward(self, x):
        out = torch.relu(self.bn1(self.conv1(x)))
        out = self.layer1(out)
        out = self.layer2(out)
        out = self.layer3(out)
        out = self.layer4(out)
        out = nn.functional.avg_pool2d(out, 4)
        out = out.view(out.size(0), -1)
        out = self.linear(out)
        return out

def ResNet18():
    return ResNet(BasicBlock, [2, 2, 2, 2])

# 创建 ResNet 模型
resnet_model = ResNet18().to(device)
print(resnet_model)
```

- **定义训练器Trainer**

> 封装好训练评估函数以及可视化测试函数

```python
class ModelTrainer:
    """模型训练器类，封装训练、评估和可视化功能"""

    def __init__(self, model, device=None):
        """
        初始化训练器
        Args:
            model: 要训练的模型
            device: 计算设备，如果为None则自动选择
        """
        self.model = model
        self.device = device if device else torch.device('cuda' if torch.cuda.is_available() el
        print(f"Using device: {device}")
        self.model = self.model.to(self.device)

        # 训练历史记录
        self.train_losses = []
        self.train_accs = []
        self.test_accs = []
        self.test_precisions = []
        self.test_recalls = []
        self.test_f1s = []

        # 最佳模型记录
        self.best_test_acc = 0
        self.best_epoch = -1

    def train(self, train_loader, test_loader, epochs=50,
              optimizer_name='adam', lr=0.001, weight_decay=0,
              scheduler_type='cosine', class_names=None, verbose=True):
        """
        训练模型
        Args:
            train_loader: 训练数据加载器
            test_loader: 测试数据加载器
            epochs: 训练轮数
            optimizer_name: 优化器名称 ('adam', 'sgd', 'rmsprop')
            lr: 学习率
            weight_decay: 权重衰减
            scheduler_type: 学习率调度器类型 ('cosine', 'step', 'none')
            class_names: 类别名称列表，用于评估时打印
            verbose: 是否打印详细信息
        Returns:
```

```
        self: 返回自身，支持链式调用
    """
    criterion = nn.CrossEntropyLoss()

    # 选择优化器
    optimizer = self._get_optimizer(optimizer_name, lr, weight_decay)

    # 选择学习率调度器
    scheduler = self._get_scheduler(optimizer, scheduler_type, epochs)

    for epoch in range(epochs):
        # 训练一个epoch
        train_loss, train_acc = self._train_epoch(
            train_loader, criterion, optimizer, epoch, epochs, verbose
        )

        # 在测试集上评估
        test_metrics = self.evaluate(test_loader, class_names=class_names, verbose=False)

        # 记录历史
        self.train_losses.append(train_loss)
        self.train_accs.append(train_acc)
        self.test_accs.append(test_metrics['accuracy'])
        self.test_precisions.append(test_metrics['precision'])
        self.test_recalls.append(test_metrics['recall'])
        self.test_f1s.append(test_metrics['f1'])

        # 保存最佳模型
        if test_metrics['accuracy'] > self.best_test_acc:
            self.best_test_acc = test_metrics['accuracy']
            self.best_epoch = epoch + 1
            self._save_best_model()

        # 更新学习率
        if scheduler:
            scheduler.step()

        # 打印进度
        if verbose:
            current_lr = optimizer.param_groups[0]['lr']
            print(f'Epoch {epoch+1}/{epochs}: '
                  f'Train Loss: {train_loss:.4f}, Train Acc: {train_acc:.2f}%, '
                  f'Test Acc: {test_metrics["accuracy"]:.2f}%, LR: {current_lr:.6f}')
```

```
    # 训练结束，恢复最佳模型
    self._restore_best_model()

    if verbose:
        print(f"\n训练完成！最佳测试准确率: {self.best_test_acc:.2f}% (Epoch {self.best_epoch

    return self

def _train_epoch(self, train_loader, criterion, optimizer, epoch, total_epochs, verbose):
    """训练一个epoch"""
    self.model.train()
    running_loss = 0.0
    correct = 0
    total = 0

    iterator = tqdm(train_loader, desc=f'Epoch {epoch+1}/{total_epochs}') if verbose else tr

    for inputs, targets in iterator:
        inputs, targets = inputs.to(self.device), targets.to(self.device)

        optimizer.zero_grad()
        outputs = self.model(inputs)
        loss = criterion(outputs, targets)
        loss.backward()
        optimizer.step()

        running_loss += loss.item()
        _, predicted = outputs.max(1)
        total += targets.size(0)
        correct += predicted.eq(targets).sum().item()

    train_loss = running_loss / len(train_loader)
    train_acc = 100. * correct / total

    return train_loss, train_acc

def evaluate(self, test_loader, class_names=None, verbose=True, num_examples=5):
    """
    评估模型
    Args:
        test_loader: 测试数据加载器
        class_names: 类别名称列表
```

```
        verbose: 是否打印详细信息
        num_examples: 打印的测试案例数量
    Returns:
        dict: 包含各种评估指标的字典
    """
    self.model.eval()

    all_preds = []
    all_targets = []
    all_images = []

    with torch.no_grad():
        for inputs, targets in test_loader:
            inputs, targets = inputs.to(self.device), targets.to(self.device)
            outputs = self.model(inputs)
            _, predicted = outputs.max(1)

            all_preds.extend(predicted.cpu().numpy())
            all_targets.extend(targets.cpu().numpy())

            # 保存前num_examples张图片用于展示
            if len(all_images) < num_examples:
                remaining = num_examples - len(all_images)
                all_images.extend(inputs[:remaining].cpu())

    # 转换为numpy数组
    all_preds = np.array(all_preds)
    all_targets = np.array(all_targets)

    # 计算各项指标
    from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score, co

    accuracy = accuracy_score(all_targets, all_preds) * 100
    precision = precision_score(all_targets, all_preds, average='macro', zero_division=0) *
    recall = recall_score(all_targets, all_preds, average='macro', zero_division=0) * 100
    f1 = f1_score(all_targets, all_preds, average='macro', zero_division=0) * 100

    # 每类的精确率和召回率
    per_class_precision = precision_score(all_targets, all_preds, average=None, zero_divisi
    per_class_recall = recall_score(all_targets, all_preds, average=None, zero_division=0)

    metrics = {
        'accuracy': accuracy,
```

```
        'precision': precision,
        'recall': recall,
        'f1': f1,
        'per_class_precision': per_class_precision,
        'per_class_recall': per_class_recall,
        'confusion_matrix': confusion_matrix(all_targets, all_preds),
        'predictions': all_preds,
        'targets': all_targets,
        'images': all_images
    }

    if verbose:
        self._print_metrics(metrics, class_names, num_examples)

    return metrics

def _print_metrics(self, metrics, class_names=None, num_examples=5):
    """打印评估指标和可视化结果"""
    print("\n" + "="*60)
    print("模型评估结果")
    print("="*60)
    print(f"准确率 (Accuracy): {metrics['accuracy']:.2f}%")
    print(f"宏平均精确率 (Macro Precision): {metrics['precision']:.2f}%")
    print(f"宏平均召回率 (Macro Recall): {metrics['recall']:.2f}%")
    print(f"宏平均F1分数 (Macro F1): {metrics['f1']:.2f}%")

    if class_names:
        print("\n各类别详细指标:")
        print("-"*60)
        print(f"{'类别':<12} {'精确率':<10} {'召回率':<10}")
        print("-"*60)
        for i, name in enumerate(class_names):
            print(f"{name:<12} {metrics['per_class_precision'][i]:<10.2f}% {metrics['per_cla

    # 混淆矩阵
    cm = metrics['confusion_matrix']
    print("\n混淆矩阵:")
    print("-"*60)
    if class_names and len(class_names) <= 10:
        print("     " + " ".join([f"{name[:3]:>4}" for name in class_names]))
        for i in range(len(class_names)):
            print(f"{class_names[i][:3]:>4} " + " ".join([f"{cm[i,j]:>4}" for j in range(le
    else:
```

```
        print(cm)

    # 可视化测试案例
    if class_names and metrics['images']:
        self._visualize_predictions(metrics, class_names, num_examples)

def _visualize_predictions(self, metrics, class_names, num_examples):
    """可视化预测结果"""
    print("\n" + "="*60)
    print(f"测试案例展示 (共{num_examples}张)")
    print("="*60)

    fig, axes = plt.subplots(1, num_examples, figsize=(15, 3))
    if num_examples == 1:
        axes = [axes]

    # 注意：这里需要知道mean和std，可以通过参数传入
    # 或者从数据加载器获取，这里使用CIFAR-10的默认值
    mean = np.array([0.4914, 0.4822, 0.4465])
    std = np.array([0.2023, 0.1994, 0.2010])

    for i in range(num_examples):
        img = metrics['images'][i]
        img = img.numpy().transpose(1, 2, 0)
        img = img * std + mean  # 反归一化
        img = np.clip(img, 0, 1)

        axes[i].imshow(img)
        true_label = class_names[metrics['targets'][i]]
        pred_label = class_names[metrics['predictions'][i]]
        color = 'green' if metrics['targets'][i] == metrics['predictions'][i] else 'red'
        axes[i].set_title(f'True: {true_label}\nPred: {pred_label}', fontsize=10, color=colo
        axes[i].axis('off')

    plt.tight_layout()
    plt.show()

def plot_history(self):
    """绘制训练历史曲线"""
    fig, axes = plt.subplots(1, 3, figsize=(15, 4))

    # 损失曲线
    axes[0].plot(self.train_losses, label='Train Loss', color='blue')
```

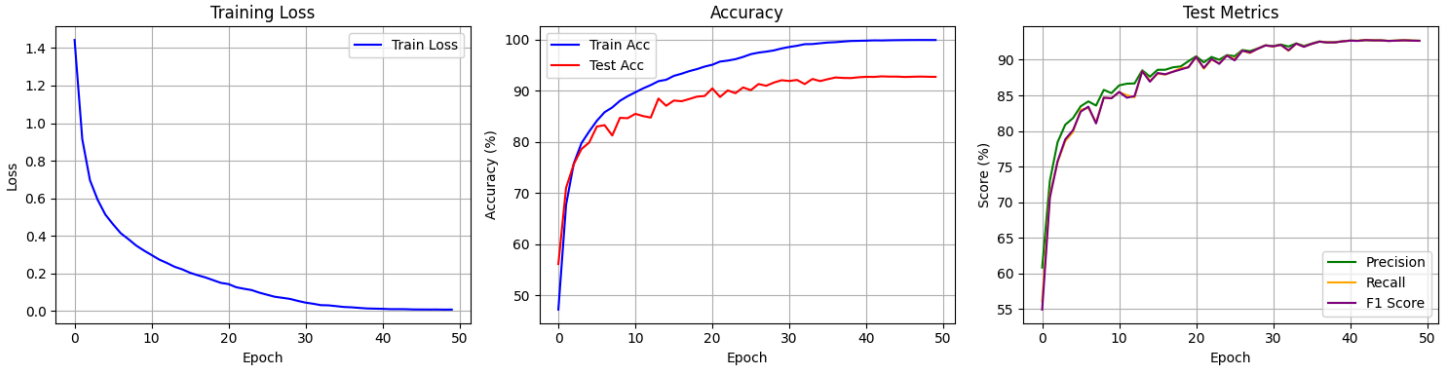
```
    axes[0].set_xlabel('Epoch')
    axes[0].set_ylabel('Loss')
    axes[0].set_title('Training Loss')
    axes[0].legend()
    axes[0].grid(True)

    # 准确率曲线
    axes[1].plot(self.train_accs, label='Train Acc', color='blue')
    axes[1].plot(self.test_accs, label='Test Acc', color='red')
    axes[1].set_xlabel('Epoch')
    axes[1].set_ylabel('Accuracy (%)')
    axes[1].set_title('Accuracy')
    axes[1].legend()
    axes[1].grid(True)

    # 其他指标曲线
    axes[2].plot(self.test_precisions, label='Precision', color='green')
    axes[2].plot(self.test_recalls, label='Recall', color='orange')
    axes[2].plot(self.test_f1s, label='F1 Score', color='purple')
    axes[2].set_xlabel('Epoch')
    axes[2].set_ylabel('Score (%)')
    axes[2].set_title('Test Metrics')
    axes[2].legend()
    axes[2].grid(True)

    plt.tight_layout()
    plt.show()

def _get_optimizer(self, optimizer_name, lr, weight_decay):
    """获取优化器"""
    if optimizer_name.lower() == 'adam':
        return optim.Adam(self.model.parameters(), lr=lr, weight_decay=weight_decay)
    elif optimizer_name.lower() == 'sgd':
        return optim.SGD(self.model.parameters(), lr=lr, momentum=0.9, weight_decay=weight_d
    elif optimizer_name.lower() == 'rmsprop':
        return optim.RMSprop(self.model.parameters(), lr=lr, weight_decay=weight_decay)
    else:
        raise ValueError(f"不支持的优化器: {optimizer_name}")

def _get_scheduler(self, optimizer, scheduler_type, epochs):
    """获取学习率调度器"""
    if scheduler_type.lower() == 'cosine':
        return optim.lr_scheduler.CosineAnnealingLR(optimizer, T_max=epochs)
```

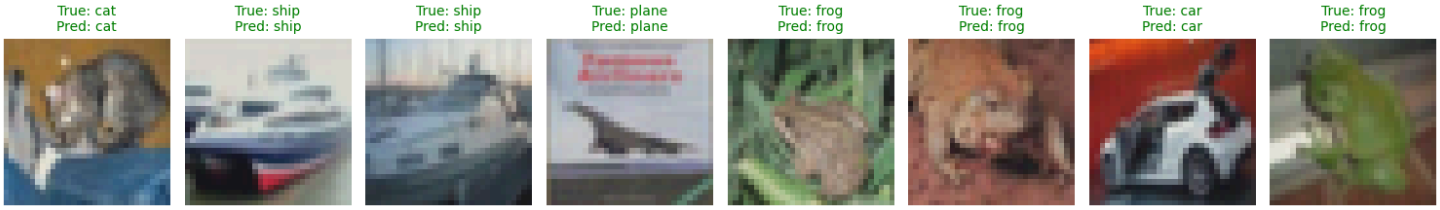
```
        elif scheduler_type.lower() == 'step':
            return optim.lr_scheduler.StepLR(optimizer, step_size=epochs//3, gamma=0.1)
        else:
            return None

    def _save_best_model(self):
        """保存最佳模型"""
        self.best_model_state = {k: v.cpu().clone() for k, v in self.model.state_dict().items()}

    def _restore_best_model(self):
        """恢复最佳模型"""
        if hasattr(self, 'best_model_state'):
            self.model.load_state_dict(self.best_model_state)
            self.model.to(self.device)

    def save_model(self, path):
        """保存模型到文件"""
        torch.save({
            'model_state_dict': self.model.state_dict(),
            'train_losses': self.train_losses,
            'train_accs': self.train_accs,
            'test_accs': self.test_accs,
            'test_precisions': self.test_precisions,
            'test_recalls': self.test_recalls,
            'test_f1s': self.test_f1s,
            'best_test_acc': self.best_test_acc,
            'best_epoch': self.best_epoch
        }, path)
        print(f"模型已保存到: {path}")

    def load_model(self, path):
        """从文件加载模型"""
        checkpoint = torch.load(path)
        self.model.load_state_dict(checkpoint['model_state_dict'])
        self.train_losses = checkpoint.get('train_losses', [])
        self.train_accs = checkpoint.get('train_accs', [])
        self.test_accs = checkpoint.get('test_accs', [])
        self.test_precisions = checkpoint.get('test_precisions', [])
        self.test_recalls = checkpoint.get('test_recalls', [])
        self.test_f1s = checkpoint.get('test_f1s', [])
        self.best_test_acc = checkpoint.get('best_test_acc', 0)
        self.best_epoch = checkpoint.get('best_epoch', -1)
        print(f"模型已从: {path} 加载")
```

```python
    def get_model(self):
        """返回模型"""
        return self.model
```

- **示例结果展示**

```python
data_loader = CIFAR10DataLoader(
        data_path='./xjtu-ml-class/exp/data/cifar-10-python',
        batch_size=128,
        num_workers=2
    )

# 2. 创建模型（这里以ResNet18为例）
Res18_model = ResNet18()  # 假设你已经定义了ResNet18

# 3. 创建训练器
trainer_res = ModelTrainer(Res18_model, device=device)

# 4. 训练模型
trainer_res.train(
    train_loader=data_loader.get_train_loader(),
    test_loader=data_loader.get_test_loader(),
    epochs=50,
    optimizer_name='sgd',
    lr=0.01,
    weight_decay=5e-4,
    scheduler_type='cosine',
    class_names=data_loader.classes,
    verbose=True
)

# 5. 绘制训练曲线
trainer_res.plot_history()

# 6. 最终评估
metrics = trainer_res.evaluate(
    test_loader=data_loader.get_test_loader(),
    class_names=data_loader.classes,
    verbose=True,
    num_examples=8
)

# 7. 保存模型
trainer_res.save_model('./Res18_rl1_sgd_batch128.pth')
```


Training Loss
Train Loss
Loss
Epoch
Accuracy
Train Acc
Test Acc
Accuracy (%)
Epoch
Test Metrics
Precision
Recall
F1 Score
Score (%)
Epoch


True: cat
Pred: cat
True: ship
Pred: ship
True: ship
Pred: ship
True: plane
Pred: plane
True: frog
Pred: frog
True: frog
Pred: frog
True: car
Pred: car
True: frog
Pred: frog

- **输出日志**

```
============================================================
模型评估结果
============================================================
准确率 (Accuracy): 92.78%
宏平均精确率 (Macro Precision): 92.77%
宏平均召回率 (Macro Recall): 92.78%
宏平均F1分数 (Macro F1): 92.77%

各类别详细指标:
------------------------------------------------------------
类别            精确率         召回率
------------------------------------------------------------
plane        92.53     % 94.10     %
car          96.52     % 97.10     %
bird         89.20     % 90.00     %
cat          87.11     % 84.50     %
deer         91.44     % 94.00     %
dog          89.63     % 88.20     %
frog         94.69     % 94.50     %
horse        95.11     % 95.40     %
ship         96.45     % 95.00     %
truck        95.00     % 95.00     %

混淆矩阵:
------------------------------------------------------------
      pla  car  bir  cat  dee  dog  fro  hor  shi  tru
 pla  941    3   17    7    6    0    1    1   17    7
 car    5  971    0    0    0    0    0    0    2   22
 bir   18    0  900   14   25   15   19    8    1    0
 cat    6    1   25  845   24   61   19   11    4    4
 dee    2    1   19    8  940   11    8   10    0    1
 dog    3    1   19   61   15  882    0   16    2    1
 fro    4    0   18   21    4    4  945    2    1    1
 hor    7    0    4    4   14   11    3  954    0    3
 shi   23    4    4    4    0    0    3    1  950   11
 tru    8   25    3    6    0    0    0    0    8  950

============================================================
测试案例展示 (共8张)
============================================================
```

> 由于训练了多个模型 一一展示过于冗杂，参加压缩包exp2的ml-exp2-img.ipynb文件中包含了训练结果的所有日志与可视化结果。

**对比试验**

| model | batch | epoch | optimizer | lr | acc | recall | precision | f1-score |
|---|---|---|---|---|---|---|---|---|
| resnet18 | 128 | 50 | sgd | 0.01 | 92.78 | 92.78 | 92.77 | 92.77 |
| vgg | 128 | 50 | sgd | 0.01 | 90.30 | 90.30 | 90.36 | 90.32 |
| resnet18 | 128 | 20 | adam | 0.001 | 80.72 | 80.63 | 80.74 | 80.68 |
| vgg | 128 | 50 | sgd | 0.005 | 91.59 | 91.59 | 91.59 | 91.59 |
| resnet18 | 256 | 50 | sgd | 0.001 | 93.29 | 93.28 | 93.29 | 93.28 |

**实验结果分析**

**1. 模型对比**

- ResNet18 整体优于 VGG，同条件下领先约 2.5%
- ResNet18 最佳 93.29% vs VGG 最佳 91.59%

**2. 优化器影响**

- SGD 效果显著优于 Adam
- ResNet18 上 SGD (93.29%) 比 Adam (80.72%) 高 12.6%

**3. 学习率影响**

- 较小学习率效果更好：lr=0.001 > 0.01（ResNet18）
- VGG 上 lr=0.005 > 0.01

**4. 批次大小**

- batch=256 略优于 batch=128（93.29% vs 92.78%）

**理论分析**：从理论角度分析，学习率直接影响参数更新的步长，较小学习率（0.001-0.005）使损失函数收敛更平稳，避免在最优解附近震荡，因此效果优于较大学习率（0.01）。Batch size 影响梯度估计的准确性，较大 batch（256）可降低梯度方差，使训练更稳定，实验显示略优于 128。优化器方面，SGD 依靠真实梯度方向更新参数，稳态收敛精度高；Adam 虽收敛速度快，但自适应学习率可能导致泛化能力下降，实验中 SGD 比 Adam 高出 12.6%，证明此分类任务更适合 SGD。模型结构上，ResNet18 引入残差连接，有效缓解梯度消失问题，深层网络优化更充分；VGG 结构较浅且无跳跃连接，特征提取

能力有限，因此 ResNet18 在各配置下均优于 VGG（领先约 2.5%）。综上，采用 ResNet18 + SGD + 较小学习率 + 较大 batch 可获得最佳效果。

## 时序预测

- **定义时序数据集类**

```
class TimeSeriesDataset(Dataset):
    """时间序列数据集类"""
    def __init__(self, data, input_len, output_len):
        if data.ndim == 1:
            data = data.reshape(-1, 1)

        self.data = torch.FloatTensor(data)
        self.input_len = input_len
        self.output_len = output_len
        self.n_features = data.shape[1]

        # 创建样本
        self.X = []
        self.y = []
        for i in range(len(data) - input_len - output_len + 1):
            self.X.append(data[i:i+input_len])
            self.y.append(data[i+input_len:i+input_len+output_len])

        self.X = torch.FloatTensor(np.array(self.X))
        self.y = torch.FloatTensor(np.array(self.y))

    def __len__(self):
        return len(self.X)

    def __getitem__(self, idx):
        return self.X[idx], self.y[idx]
```

- **定义数据预处理类**

```python
class ECGDataLoader:
    """ECG数据加载器 - 一键加载和预处理（支持单文件或多文件）"""

    def __init__(self, filepaths, input_len=50, output_len=10,
                 train_ratio=0.7, val_ratio=0.15, batch_size=64):
        """
        一键加载ECG数据并创建数据加载器

        Args:
            filepaths: 文件路径，可以是字符串（单文件）或列表（多文件）
            input_len: 输入序列长度
            output_len: 输出序列长度（预测步数）
            train_ratio: 训练集比例
            val_ratio: 验证集比例
            batch_size: 批次大小
        """
        self.input_len = input_len
        self.output_len = output_len
        self.batch_size = batch_size
        self.scaler = StandardScaler()

        # 加载并预处理数据
        self._load_data(filepaths)
        self._prepare_datasets(train_ratio, val_ratio)
        self._create_dataloaders()

    def _load_data(self, filepaths):
        """加载单文件或多文件数据"""
        # 统一转换为列表
        if isinstance(filepaths, str):
            filepaths = [filepaths]

        all_data = []
        for filepath in filepaths:
            data = []
            with open(filepath, 'r') as f:
                for line in f:
                    line = line.strip()
                    if line:
                        try:
                            data.append(float(line))
                        except:
                            continue
```

```
        if data:
            data_array = np.array(data)
            all_data.extend(data_array)
            print(f"加载 {Path(filepath).name}: {len(data_array)} 个数据点")
        else:
            print(f"警告: {filepath} 无有效数据")

    if not all_data:
        raise ValueError("没有加载到任何数据")

    self.raw_data = np.array(all_data)
    print(f"\n总数据点: {len(self.raw_data)}")
    print(f"数据范围: [{self.raw_data.min():.2f}, {self.raw_data.max():.2f}]")

def _prepare_datasets(self, train_ratio, val_ratio):
    """准备数据集"""
    # 标准化
    data_scaled = self.scaler.fit_transform(self.raw_data.reshape(-1, 1)).flatten()

    # 划分
    n = len(data_scaled)
    train_end = int(n * train_ratio)
    val_end = int(n * (train_ratio + val_ratio))

    train_data = data_scaled[:train_end]
    val_data = data_scaled[train_end:val_end]
    test_data = data_scaled[val_end:]

    # 创建数据集
    self.train_dataset = TimeSeriesDataset(train_data, self.input_len, self.output_len)
    self.val_dataset = TimeSeriesDataset(val_data, self.input_len, self.output_len)
    self.test_dataset = TimeSeriesDataset(test_data, self.input_len, self.output_len)

    print(f"\n数据集划分:")
    print(f"  训练样本: {len(self.train_dataset)}")
    print(f"  验证样本: {len(self.val_dataset)}")
    print(f"  测试样本: {len(self.test_dataset)}")

def _create_dataloaders(self):
    """创建数据加载器"""
    self.train_loader = DataLoader(self.train_dataset, batch_size=self.batch_size, shuffle=T
    self.val_loader = DataLoader(self.val_dataset, batch_size=self.batch_size, shuffle=False
```

```
        self.test_loader = DataLoader(self.test_dataset, batch_size=self.batch_size, shuffle=Fa

    def inverse_transform(self, data):
        """反标准化"""
        return self.scaler.inverse_transform(data.reshape(-1, 1)).flatten()
```

- **实现LSTM**

```python
class LSTMCell(nn.Module):
    """单个LSTM单元"""
    def __init__(self, input_size, hidden_size):
        super(LSTMCell, self).__init__()
        self.input_size = input_size
        self.hidden_size = hidden_size

        # 输入门、遗忘门、输出门、细胞状态更新的权重和偏置
        self.W_ih = nn.Linear(input_size, 4 * hidden_size, bias=True)
        self.W_hh = nn.Linear(hidden_size, 4 * hidden_size, bias=True)

    def forward(self, x, state):
        h, c = state

        # 线性变换
        gates = self.W_ih(x) + self.W_hh(h)

        # 分割成4个门
        i_gate, f_gate, g_gate, o_gate = gates.chunk(4, dim=1)

        # 激活函数
        i = torch.sigmoid(i_gate)  # 输入门
        f = torch.sigmoid(f_gate)  # 遗忘门
        g = torch.tanh(g_gate)     # 候选细胞状态
        o = torch.sigmoid(o_gate)  # 输出门

        # 更新细胞状态和隐藏状态
        c_next = f * c + i * g
        h_next = o * torch.tanh(c_next)

        return h_next, c_next

class LSTMTimeSeries(nn.Module):
    """从零实现的多层LSTM模型"""
    def __init__(self, input_size, hidden_size, num_layers, output_size, dropout=0.2):
        super(LSTMTimeSeries, self).__init__()
        self.hidden_size = hidden_size
        self.num_layers = num_layers
        self.dropout_rate = dropout

        # 创建多层LSTM单元
        self.cells = nn.ModuleList()
        for i in range(num_layers):
```

```
        in_size = input_size if i == 0 else hidden_size
        self.cells.append(LSTMCell(in_size, hidden_size))

    # Dropout层
    self.dropout = nn.Dropout(dropout)

    # 输出层
    self.fc = nn.Linear(hidden_size, output_size)

def forward(self, x):
    batch_size, seq_len, _ = x.shape
    device = x.device

    # 初始化隐藏状态和细胞状态
    h = [torch.zeros(batch_size, self.hidden_size).to(device) for _ in range(self.num_layers
    c = [torch.zeros(batch_size, self.hidden_size).to(device) for _ in range(self.num_layers

    # 遍历时间步
    for t in range(seq_len):
        x_t = x[:, t, :]

        # 逐层更新
        for layer in range(self.num_layers):
            if layer == 0:
                input_t = x_t
            else:
                input_t = h[layer-1]
                if t < seq_len - 1:
                    input_t = self.dropout(input_t)

            h[layer], c[layer] = self.cells[layer](input_t, (h[layer], c[layer]))

        if t == seq_len - 1:
            last_output = h[-1]

    last_output = self.dropout(last_output)
    output = self.fc(last_output)

    return output
```

- **实现GRU**

```python
class GRUCell(nn.Module):
    """单个GRU单元"""
    def __init__(self, input_size, hidden_size):
        super(GRUCell, self).__init__()
        self.input_size = input_size
        self.hidden_size = hidden_size

        # 更新门、重置门、候选隐藏状态的权重
        self.W_ih = nn.Linear(input_size, 3 * hidden_size, bias=True)
        self.W_hh = nn.Linear(hidden_size, 3 * hidden_size, bias=True)

    def forward(self, x, h):
        # 线性变换
        gates_x = self.W_ih(x)
        gates_h = self.W_hh(h)

        # 分割成3个门
        z_gate, r_gate, n_gate = gates_x.chunk(3, dim=1)
        z_h, r_h, n_h = gates_h.chunk(3, dim=1)

        # 更新门和重置门
        z = torch.sigmoid(z_gate + z_h)
        r = torch.sigmoid(r_gate + r_h)

        # 候选隐藏状态
        n = torch.tanh(n_gate + r * n_h)

        # 更新隐藏状态
        h_next = (1 - z) * n + z * h

        return h_next

class GRUTimeSeries(nn.Module):
    """从零实现的多层GRU模型"""
    def __init__(self, input_size, hidden_size, num_layers, output_size, dropout=0.2):
        super(GRUTimeSeries, self).__init__()
        self.hidden_size = hidden_size
        self.num_layers = num_layers
        self.dropout_rate = dropout

        # 创建多层GRU单元
        self.cells = nn.ModuleList()
        for i in range(num_layers):
```

```
            in_size = input_size if i == 0 else hidden_size
            self.cells.append(GRUCell(in_size, hidden_size))

        # Dropout层
        self.dropout = nn.Dropout(dropout)

        # 输出层
        self.fc = nn.Linear(hidden_size, output_size)

    def forward(self, x):
        batch_size, seq_len, _ = x.shape
        device = x.device

        # 初始化隐藏状态
        h = [torch.zeros(batch_size, self.hidden_size).to(device) for _ in range(self.num_layers

        # 遍历时间步
        for t in range(seq_len):
            x_t = x[:, t, :]

            # 逐层更新
            for layer in range(self.num_layers):
                if layer == 0:
                    input_t = x_t
                else:
                    input_t = h[layer-1]
                    if t < seq_len - 1:
                        input_t = self.dropout(input_t)

                h[layer] = self.cells[layer](input_t, h[layer])

            if t == seq_len - 1:
                last_output = h[-1]

        last_output = self.dropout(last_output)
        output = self.fc(last_output)

        return output
```

- **定义训练器Trainer**

```python
class TimeSeriesTrainer:
    """时间序列预测训练器类"""

    def __init__(self, model, device=None):
        """
        初始化训练器
        Args:
            model: 要训练的模型
            device: 计算设备，如果为None则自动选择
        """
        self.model = model
        self.device = device if device else torch.device('cuda' if torch.cuda.is_available() el
        print(f"Using device: {self.device}")
        self.model = self.model.to(self.device)

        # 初始化记录
        self.reset()

    def reset(self):
        """重置训练历史记录（不重置模型参数）"""
        self.train_losses = []
        self.val_losses = []
        self.test_maes = []
        self.test_rmses = []
        self.step_maes_history = []

        self.best_val_loss = float('inf')
        self.best_epoch = -1
        self.best_model_state = None

        print("训练历史已重置")
        return self

    def reset_model(self):
        """重置模型参数（重新初始化）"""
        def init_weights(m):
            if isinstance(m, nn.Linear):
                nn.init.xavier_uniform_(m.weight)
                if m.bias is not None:
                    nn.init.zeros_(m.bias)
            elif isinstance(m, nn.LSTM):
                for name, param in m.named_parameters():
                    if 'weight_ih' in name:
```

```
                nn.init.xavier_uniform_(param)
            elif 'weight_hh' in name:
                nn.init.orthogonal_(param)
            elif 'bias' in name:
                nn.init.zeros_(param)
        elif isinstance(m, nn.GRU):
            for name, param in m.named_parameters():
                if 'weight_ih' in name:
                    nn.init.xavier_uniform_(param)
                elif 'weight_hh' in name:
                    nn.init.orthogonal_(param)
                elif 'bias' in name:
                    nn.init.zeros_(param)

    self.model.apply(init_weights)
    self.reset()
    print("模型参数已重新初始化")
    return self

def train(self, train_loader, val_loader, epochs=100,
          optimizer_name='adam', lr=0.001, weight_decay=0,
          scheduler_type='cosine', early_stopping_patience=None,
          verbose=True, reset_history=True):
    """
    训练模型
    """
    if reset_history:
        self.reset()

    criterion = nn.MSELoss()
    optimizer = self._get_optimizer(optimizer_name, lr, weight_decay)
    scheduler = self._get_scheduler(optimizer, scheduler_type, epochs)

    patience_counter = 0

    for epoch in range(epochs):
        # 训练一个epoch
        train_loss = self._train_epoch(train_loader, criterion, optimizer, epoch, epochs, ve

        # 验证
        val_metrics = self.evaluate(val_loader, verbose=False)

        # 记录历史
```

```
        self.train_losses.append(train_loss)
        self.val_losses.append(val_metrics['MAE'])
        self.test_maes.append(val_metrics['MAE'])
        self.test_rmses.append(val_metrics['RMSE'])
        self.step_maes_history.append(val_metrics['step_MAE'])

        # 保存最佳模型
        if val_metrics['MAE'] < self.best_val_loss:
            self.best_val_loss = val_metrics['MAE']
            self.best_epoch = epoch + 1
            self._save_best_model()
            patience_counter = 0
        else:
            patience_counter += 1

        # 早停检查
        if early_stopping_patience and patience_counter >= early_stopping_patience:
            if verbose:
                print(f"\n早停触发于 epoch {epoch+1}")
            break

        # 更新学习率
        if scheduler:
            if scheduler_type.lower() == 'plateau':
                scheduler.step(val_metrics['MAE'])
            else:
                scheduler.step()

        # 打印进度
        if verbose and (epoch + 1) % 10 == 0:
            current_lr = optimizer.param_groups[0]['lr']
            print(f'Epoch {epoch+1}/{epochs}: '
                  f'Train Loss: {train_loss:.6f}, '
                  f'Val MAE: {val_metrics["MAE"]:.4f}, '
                  f'Val RMSE: {val_metrics["RMSE"]:.4f}, '
                  f'LR: {current_lr:.6f}')

    # 训练结束，恢复最佳模型
    self._restore_best_model()

    if verbose:
        print(f"\n训练完成！最佳验证MAE: {self.best_val_loss:.4f} (Epoch {self.best_epoch})")
```

```
        return self

    def _train_epoch(self, train_loader, criterion, optimizer, epoch, total_epochs, verbose):
        """训练一个epoch"""
        self.model.train()
        running_loss = 0.0

        iterator = tqdm(train_loader, desc=f'Epoch {epoch+1}/{total_epochs}') if verbose else tr

        for X_batch, y_batch in iterator:
            X_batch = X_batch.to(self.device)
            y_batch = y_batch.to(self.device)
            if y_batch.dim() == 3 and y_batch.shape[-1] == 1:
              y_batch = y_batch.squeeze(-1)

            optimizer.zero_grad()
            y_pred = self.model(X_batch)
            loss = criterion(y_pred, y_batch)
            loss.backward()
            torch.nn.utils.clip_grad_norm_(self.model.parameters(), max_norm=1.0)
            optimizer.step()

            running_loss += loss.item() * X_batch.size(0)

            if verbose:
                iterator.set_postfix({'loss': loss.item()})

        train_loss = running_loss / len(train_loader.dataset)
        return train_loss

    def evaluate(self, dataloader, scaler=None, verbose=False):
        """评估模型"""
        self.model.eval()

        predictions = []
        targets = []

        with torch.no_grad():
            for X_batch, y_batch in dataloader:
                X_batch = X_batch.to(self.device)
                y_batch = y_batch.to(self.device)
                if y_batch.dim() == 3 and y_batch.shape[-1] == 1:
                  y_batch = y_batch.squeeze(-1)
```

```
            y_pred = self.model(X_batch)

            predictions.append(y_pred.cpu().numpy())
            targets.append(y_batch.cpu().numpy())

    predictions = np.concatenate(predictions, axis=0)
    targets = np.concatenate(targets, axis=0)

    # 反标准化
    if scaler:
        n_samples, n_steps = predictions.shape
        predictions = scaler.inverse_transform(predictions.reshape(-1, 1)).reshape(n_samples
        targets = scaler.inverse_transform(targets.reshape(-1, 1)).reshape(n_samples, n_step

    # 计算指标
    mae = mean_absolute_error(targets.flatten(), predictions.flatten())
    rmse = np.sqrt(mean_squared_error(targets.flatten(), predictions.flatten()))

    step_mae = []
    step_rmse = []
    for step in range(targets.shape[1]):
        step_mae.append(mean_absolute_error(targets[:, step], predictions[:, step]))
        step_rmse.append(np.sqrt(mean_squared_error(targets[:, step], predictions[:, step]))

    metrics = {
        'MAE': mae,
        'RMSE': rmse,
        'step_MAE': step_mae,
        'step_RMSE': step_rmse,
        'predictions': predictions,
        'targets': targets
    }

    if verbose:
        self._print_metrics(metrics)

    return metrics

def _print_metrics(self, metrics):
    """打印评估指标"""
    print("\n" + "="*60)
    print("模型评估结果")
    print("="*60)
```

```
        print(f"整体MAE: {metrics['MAE']:.4f}")
        print(f"整体RMSE: {metrics['RMSE']:.4f}")
        print(f"\n各步长预测结果:")
        print(f"{'步长':<8} {'MAE':<12} {'RMSE':<12}")
        print("-"*32)
        for i in range(len(metrics['step_MAE'])):
            print(f"Step {i+1:<3} {metrics['step_MAE'][i]:<12.4f} {metrics['step_RMSE'][i]:<12.4

    def predict(self, X, scaler=None):
        """对单个输入进行预测"""
        self.model.eval()

        if not isinstance(X, torch.Tensor):
            X = torch.FloatTensor(X)

        if X.dim() == 2:
            X = X.unsqueeze(0)

        X = X.to(self.device)

        with torch.no_grad():
            predictions = self.model(X)
            predictions = predictions.cpu().numpy()

        if scaler:
            n_samples, n_steps = predictions.shape
            predictions = scaler.inverse_transform(predictions.reshape(-1, 1)).reshape(n_samples

        return predictions.squeeze()

    def plot_history(self):
        """绘制训练历史曲线"""
        fig, axes = plt.subplots(1, 2, figsize=(12, 4))

        # 损失曲线
        axes[0].plot(self.train_losses, label='Train Loss', color='blue')
        axes[0].set_xlabel('Epoch')
        axes[0].set_ylabel('Loss')
        axes[0].set_title('Training Loss')
        axes[0].legend()
        axes[0].grid(True)

        # 验证MAE曲线
```

```
        axes[1].plot(self.val_losses, label='Val MAE', color='red')
        axes[1].set_xlabel('Epoch')
        axes[1].set_ylabel('MAE')
        axes[1].set_title('Validation MAE')
        axes[1].legend()
        axes[1].grid(True)

        plt.tight_layout()
        plt.show()

    def plot_predictions(self, dataloader, scaler=None, num_samples=3):
        """绘制预测结果对比图"""
        metrics = self.evaluate(dataloader, scaler=scaler, verbose=False)
        predictions = metrics['predictions']
        targets = metrics['targets']

        fig, axes = plt.subplots(num_samples, 1, figsize=(10, 3*num_samples))
        if num_samples == 1:
            axes = [axes]

        for i in range(min(num_samples, len(predictions))):
            steps = range(1, len(predictions[i]) + 1)
            axes[i].plot(steps, targets[i], 'o-', label='Actual', markersize=6, linewidth=2)
            axes[i].plot(steps, predictions[i], 's--', label='Predicted', markersize=6, linewid
            axes[i].set_xlabel('Prediction Step')
            axes[i].set_ylabel('Value')
            axes[i].set_title(f'Sample {i+1}')
            axes[i].legend()
            axes[i].grid(True)

        plt.tight_layout()
        plt.show()

    def _get_optimizer(self, optimizer_name, lr, weight_decay):
        """获取优化器"""
        if optimizer_name.lower() == 'adam':
            return optim.Adam(self.model.parameters(), lr=lr, weight_decay=weight_decay)
        elif optimizer_name.lower() == 'sgd':
            return optim.SGD(self.model.parameters(), lr=lr, momentum=0.9, weight_decay=weight_
        elif optimizer_name.lower() == 'rmsprop':
            return optim.RMSprop(self.model.parameters(), lr=lr, weight_decay=weight_decay)
        else:
            raise ValueError(f"不支持的优化器: {optimizer_name}")
```

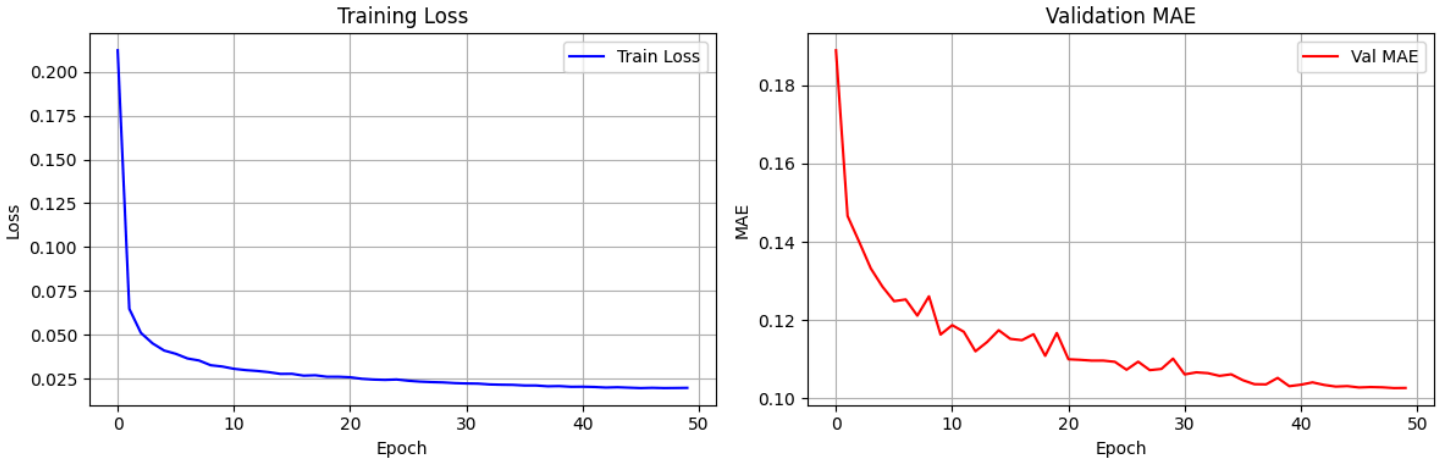
```python
def _get_scheduler(self, optimizer, scheduler_type, epochs):
    """获取学习率调度器"""
    if scheduler_type.lower() == 'cosine':
        return optim.lr_scheduler.CosineAnnealingLR(optimizer, T_max=epochs)
    elif scheduler_type.lower() == 'step':
        return optim.lr_scheduler.StepLR(optimizer, step_size=epochs//3, gamma=0.1)
    elif scheduler_type.lower() == 'plateau':
        return optim.lr_scheduler.ReduceLROnPlateau(optimizer, mode='min', patience=10, fact
    else:
        return None

def _save_best_model(self):
    """保存最佳模型"""
    self.best_model_state = {k: v.cpu().clone() for k, v in self.model.state_dict().items()

def _restore_best_model(self):
    """恢复最佳模型"""
    if hasattr(self, 'best_model_state') and self.best_model_state:
        self.model.load_state_dict(self.best_model_state)
        self.model.to(self.device)

def save_model(self, path):
    """保存模型到文件"""
    torch.save({
        'model_state_dict': self.model.state_dict(),
        'train_losses': self.train_losses,
        'val_losses': self.val_losses,
        'best_val_loss': self.best_val_loss,
        'best_epoch': self.best_epoch
    }, path)
    print(f"模型已保存到: {path}")

def load_model(self, path):
    """从文件加载模型"""
    checkpoint = torch.load(path, map_location=self.device)
    self.model.load_state_dict(checkpoint['model_state_dict'])
    self.train_losses = checkpoint.get('train_losses', [])
    self.val_losses = checkpoint.get('val_losses', [])
    self.best_val_loss = checkpoint.get('best_val_loss', float('inf'))
    self.best_epoch = checkpoint.get('best_epoch', -1)
    print(f"模型已从: {path} 加载")
```

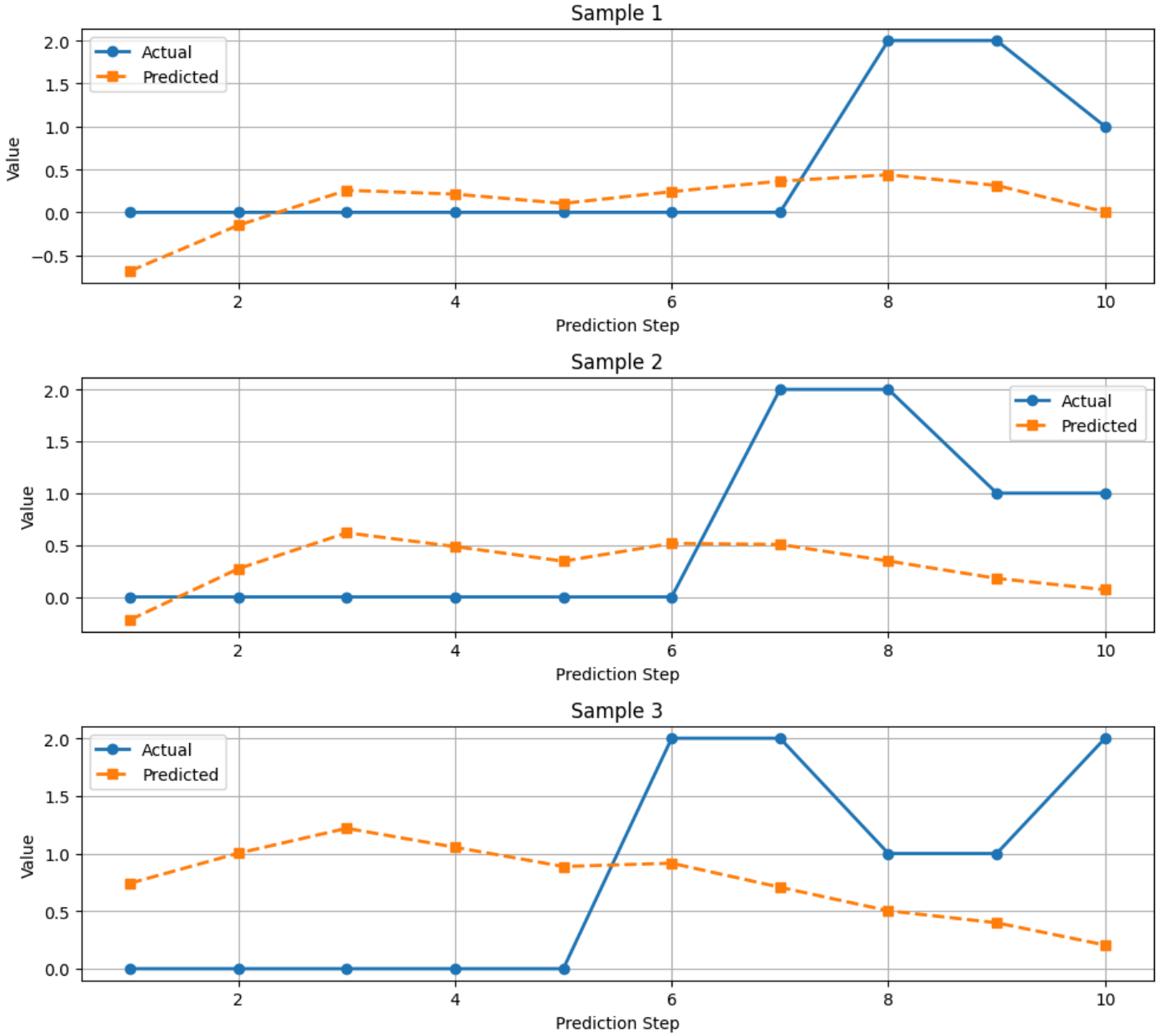
```python
    def get_model(self):
        """返回模型"""
        return self.model
```

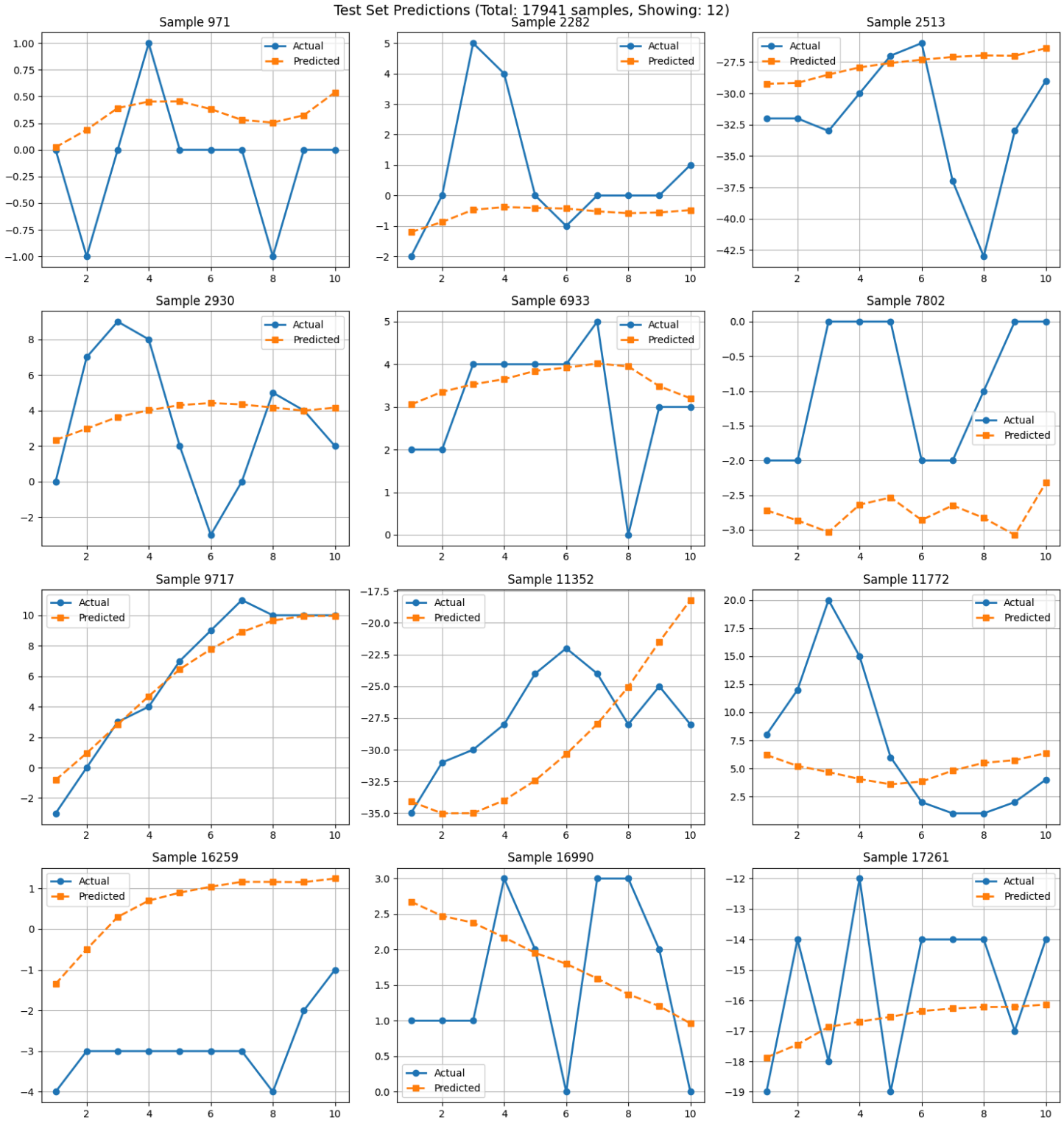
- **示例结果展示**

```python
# 1. 加载数据
data_loader = ECGDataLoader(
    filepaths=filepaths,
    input_len=50,
    output_len=10,
    batch_size=64
)
# 2. 创建模型
model = LSTMTimeSeries(
    input_size=1,        # 单变量时间序列
    hidden_size=64,
    num_layers=2,
    output_size=10,      # 预测10个点
    dropout=0.2
)
# 3. 创建训练器
trainer = TimeSeriesTrainer(model, device)
# 4. 训练模型
trainer.train(
    train_loader=data_loader.train_loader,
    val_loader=data_loader.val_loader,
    epochs=50,
    optimizer_name='adam',
    lr=0.001,
    weight_decay=1e-5,
    scheduler_type='cosine',
    early_stopping_patience=20,
    verbose=True
)
# 5. 绘制训练曲线
trainer.plot_history()
# 6. 最终评估
metrics = trainer.evaluate(
    dataloader=data_loader.test_loader,
    scaler=data_loader.scaler,
    verbose=True
)
# 7. 可视化预测结果
trainer.plot_predictions(
    dataloader=data_loader.test_loader,
    scaler=data_loader.scaler,
    num_samples=3
)
```

```
# 8. 保存模型
trainer.save_model('/content/drive/lstm_lr1_adam_batch64.pth')
```




Sample 1
Actual
Predicted
Value
Prediction Step
Sample 2
Actual
Predicted
Value
Prediction Step
Sample 3
Actual
Predicted
Value
Prediction Step


Test Set Predictions (Total: 17941 samples, Showing: 12)
Sample 971
Sample 2282
Sample 2513
Sample 2930
Sample 6933
Sample 7802
Sample 9717
Sample 11352
Sample 11772
Sample 16259
Sample 16990
Sample 17261
Actual
Predicted

- **输出日志**

```
============================================================
模型评估结果
============================================================
整体MAE: 2.6225
整体RMSE: 5.0789

各步长预测结果:
步长        MAE          RMSE
--------------------------------
Step 1   1.9191       3.3556
Step 2   2.4295       4.1838
Step 3   2.5409       4.5332
Step 4   2.5791       4.7557
Step 5   2.6354       4.9876
Step 6   2.6822       5.2374
Step 7   2.7363       5.4658
Step 8   2.8065       5.6444
Step 9   2.9002       5.8646
Step 10  2.9962       6.1313
```

**对比试验**

| model | in-out | batch | epoch | optimizer | lr | MAE | RMSE |
|---|---|---|---|---|---|---|---|
| LSTM | 50-10 | 64 | 50 | adam | 0.001 | 0.1027 | 0.1661 |
| GRU | 50-10 | 64 | 50 | adam | 0.001 | 0.1040 | 0.1735 |
| LSTM | 10-1 | 64 | 20 | adam | 0.005 | 0.0851 | 0.1265 |
| GRU | 20-5 | 64 | 20 | sgd | 0.001 | 0.2706 | 0.4645 |
| LSTM | 20-10 | 256 | 50 | adam | 0.005 | 0.1219 | 0.2124 |

**实验结果分析**

**1. 模型对比**

- LSTM 整体略优于 GRU
- 同条件下 LSTM (0.1027) vs GRU (0.1040)，差距不明显

**2. 预测任务难度**

- 单步预测（10-1）效果最好，MAE=0.0851

- 多步预测（50-10）误差明显增大，MAE=0.1027
- 累积误差是多步预测的主要挑战

**3. 优化器影响**

- Adam 效果优于 SGD
- SGD 配置下 MAE=0.2706，远高于 Adam

**4. 输入输出窗口**

- 输入 10 预测 1：效果最佳
- 输入 50 预测 10：误差适中
- 输入 20 预测 5（SGD）：效果最差

**5. 批次大小**

- batch=64（0.1027）优于 batch=256（0.1219）
- 较小批次更适合时序任务

**理论分析**:从时序预测角度，**输入输出窗口**直接影响任务难度：输出步长越长，累积误差越明显，因此单步预测效果最优。**模型结构上**，LSTM 的门控机制比 GRU 更复杂，对长期依赖的建模能力略强。**优化器方面**，Adam 的自适应学习率适合时序任务的非平稳特性，而 SGD 收敛不稳定。**批次大小**影响梯度估计，较小批次引入的噪声反而有助于时序模型跳出局部最优。综上，时序预测应优先选择 **LSTM + Adam + 单步预测 + 较小批次**。

# 总结

- 通过本次图像分类实验，我掌握了使用 PyTorch 构建 VGG 和 ResNet 等经典卷积神经网络的方法，理解了数据加载、数据增强、归一化等预处理流程对模型训练的重要性。实验中我系统调优了学习率、批次大小、优化器等超参数，深刻体会到不同超参数对模型收敛速度和泛化能力的影响：SGD 相比 Adam 在此任务上具有更高的稳态精度，较小学习率有助于损失函数平稳收敛，较大批次能降低梯度方差。此外，通过对比 ResNet 和 VGG 的结构差异，我认识到残差连接能有效缓解梯度消失问题，使深层网络优化更加充分。这些实践加深了我对深度学习训练流程和超参数调优的理解，为后续模型设计与实验提供了宝贵经验。
- 通过本次时序预测实验，我掌握了使用 PyTorch 从零实现 LSTM 和 GRU 单元的方法，理解了时间序列数据集的构建流程，包括滑动窗口采样、标准化处理以及训练/验证/测试集按时间顺序划分的原则。实验中我系统调整了输入输出窗口长度、批次大小、优化器等超参数，深刻体会到不同因素对多步预测任务的影响：输出步长越长，累积误差越明显，单步预测精度远高于多步预测；LSTM 的门控机制比 GRU 略具优势，但差距不大；Adam 优化器因其自适应学习率特性，在时序任务上表现优于 SGD；较小批次引入的梯度噪声反而有助于时序模型泛化。此外，通过对比验证集与测试集的误差差异，我认识到时序数据划分必须严格按时间顺序，避免未来信息泄露。这些实践加深

了我对循环神经网络工作原理和时序预测任务特性的理解，为后续处理更复杂的序列问题奠定了基础。

# 附录

参见压缩包exp2的ml-exp2-img.ipynb文件